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A Calendar-Aware Frequency-Decoupled Framework for Day-Ahead Substation Load Forecasting Using SHAP-Based Interpretation

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Abstract

Accurate substation-level Short-Term Load Forecasting (STLF) is essential for secure day-ahead power-system operation, yet localized demand is often affected by meteorological variation and discrete calendar shifts such as statutory holidays and makeup workdays. At this spatial scale, end-to-end forecasting models may over-smooth abrupt local changes and fail to represent peaks and valleys accurately. To address this issue, this study proposes a Calendar-Aware Frequency-Decoupled Framework (CA-FDF) for 24 h ahead substation load forecasting. The load series is decomposed by the Discrete Wavelet Transform (DWT), and the low-frequency component is tracked by a regime-aware Recursive Least Squares (RLS) filter. The residuals are then refined through explicit calendar-state encoding and day-ahead weather forecasts. A Multi-Layer Perceptron (MLP) learns latent weather representations, while SHapley Additive exPlanations (SHAP) interpret calendar- and weather-related effects. Experiments on hourly operational data from 29 anonymized substations in East China show that CA-FDF achieves a Mean Absolute Percentage Error (MAPE) of 1.92% and outperforms representative baselines under the same day-ahead setting. The results indicate that frequency-decoupled residual refinement improves localized load forecasting, with calendar-aware correction contributing the largest gain.

Keywords: short-term load forecasting; calendar-state encoding; frequency decoupling; over-smoothing

1. Introduction

The transition to sustainable power grids and the increasing frequency of extreme weather events have brought greater volatility to modern power systems [1]. Therefore, Short-Term Load Forecasting (STLF) remains an important task for economic dispatch and grid security. Although system-level aggregated load forecasting has achieved high accuracy, predicting demand at the localized distribution level remains difficult. Local loads are closely tied to specific industrial activities and severe meteorological variations [2]. In addition, the growing penetration of solar and other distributed resources further complicates net load forecasting at the distribution level [3].

Recent studies have continued to explore hybrid and decomposition-enhanced structures for short-term load forecasting. Cai et al. proposed a Variational Mode Decomposition–Gated Recurrent Unit–Temporal Convolutional Network (VMD-GRU-TCN) hybrid network to combine signal decomposition with deep temporal modeling for electrical load

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forecasting [4]. Xiong et al. further integrated VMD, deep temporal convolutional networks, and self-attention to capture multi-scale short-term load variation [5]. Chen et al. studied improved data decomposition and hybrid deep-learning models for short-term electricity load forecasting, showing that decomposition-based preprocessing remains useful for improving forecasting performance [6]. These studies indicate that hybrid forecasting pipelines are still important in practical power-system applications. A recent comprehensive review by Eren and Kucukdemiral further summarized deep-learning-based STLF studies and identified important research directions related to dataset specification, uncertainty-aware forecasting, online solutions, and practical extensions [7]. These findings suggest that further work is still needed on localized substation forecasting, explicit calendar-state modeling, reproducible experimental settings, and interpretable feature attribution.

Recent progress in STLF has been driven largely by advanced deep learning (DL) architectures that capture nonlinear temporal dependencies. Ensemble tree-based algorithms such as Extreme Gradient Boosting (XGBoost) have shown strong performance on structured tabular data [8], and optimized boosting frameworks have been proposed to reduce asymmetric forecasting deviations [9]. Meanwhile, hybrid recurrent networks, such as spatio-temporal fusion graphs, can model coupled temporal representations [10]. Optimized Convolutional Neural Network–Bidirectional Long Short-Term Memory–Attention (CNN-BiLSTM-Attention) structures have also been applied successfully to sequential grid data [11]. Recent studies have further shown that neural-network-based and transfer-learning-based frameworks can achieve strong performance in STLF under weather-sensitive and heterogeneous operating conditions [12,13].

Transformer-based models and their variants have recently become prominent because of their ability to capture long-range dependencies. In load forecasting and closely related power-system forecasting tasks, feature-temporal Transformers have been used for multivariate weather inputs [14], multi-scale hypergraph and dual-layer Transformer structures have been applied to complex grid forecasting problems [15], and hybrid Transformer-TCN-GRU models have improved thermal load forecasting [16]. Comparative studies across multiple time scales also continue to support the value of recurrent and Transformer-based architectures for load forecasting [17]. Meanwhile, patch-based Transformer models such as PatchTST have been developed for long-horizon forecasting [18], and simpler linear baselines such as DLinear remain competitive when decomposition is carefully designed [19].

Recent studies have shown more clearly that the main difficulty arises when forecasting moves from aggregated demand to local network levels. Pinheiro et al. [20] developed a forecasting framework spanning from the system level to secondary substations and showed that, at more localized and volatile nodes, historical load, calendar information, and weather variables need to be considered together. López et al. [21] examined uncommon calendar conditions and showed that a simple holiday label is often not sufficient. With a finer distinction among holidays and adjacent days, the forecasting accuracy can be improved. Cao et al. [22] discussed the same problem from an adaptation perspective and noted that external changes, especially those related to weather and holidays, may rapidly lead to concept drift and performance degradation. Similar findings were also observed in distribution-level prediction research. Huyghues-Beaufond et al. [23] showed that automatic data cleansing and handling of outliers have a clear effect on 24 h ahead forecasting, while Rubasinghe et al. [24] found that a two-stage design using decomposition is helpful for tracking peaks and valleys during periods of high short-term volatility. Related findings also appear in studies based on component decomposition and combined modeling. Massidda et al. [25] modeled total load and thermal load of communities separately, and Pramanik et al. [26] combined similar-day information with sequence modeling in buildings with high renewable penetration to capture both smooth background demand and short-term variation. Together, these studies indicate that forecasting at more localized levels depends not only on temporal dependence itself but also on data quality, calendar effects, variation at different time scales, and changing external conditions.

Existing studies have improved the accuracy of forecasting, but when data-driven models are applied to substation-level load series, there still remains a significant issue. In many cases, these models produce overly smooth predictions and fail to follow abrupt local changes well. This is partly because, under global loss functions, high-frequency variations are often averaged out to reduce the overall error. To address the issue of unstable data, researchers have tried feature separation and fusion methods [27], as well as more detailed decomposition strategies [28]. For instance, decomposed Fourier AutoRegressive Integrated Moving Average (ARIMA) has been used to model multi-frequency seasonal patterns in energy markets [29], and adaptive multi-stage wavelet denoising has been used to keep load characteristics useful while filtering noise [30]. However, previous work has usually emphasized either special-day modeling or adaptation to changing data distributions. Little attention has been paid to combining explicit calendar-state modeling with a frequency-decoupled forecasting structure under day-ahead weather constraints.

To address this issue, we propose a Calendar-Aware Frequency-Decoupled Framework (CA-FDF) for 24 h ahead substation load forecasting. The framework first stabilizes the low-frequency base of the load series and then refines the remaining residuals using calendar-state information and day-ahead meteorological inputs. This staged design is intended to keep the slow-moving structure stable while retaining sensitivity to local shifts associated with special calendar conditions and weather changes. To remain consistent with practical dispatch settings, the model uses only day-ahead weather forecasts rather than realized future weather observations.

The main contributions of this paper are as follows:

- **A staged framework to alleviate over-smoothing in substation-level forecasting:** We address a practical weakness of end-to-end forecasting models at substation level, namely their tendency to smooth out sharp local variations. By separating low-frequency base tracking from high-frequency residual correction, the proposed framework improves the representation of peaks, valleys, and rapid structural changes.
- **An explicit calendar-state representation for institutional schedule shifts:** Instead of using only routine timestamp features or a binary holiday flag, we encode Weekday, Weekend, Holiday, and Makeup as distinct calendar states. This representation method is in line with the actual effects of special holiday arrangements in China and helps the residual module distinguish the types of unusual days more effectively.
- **Validation under a strict day-ahead forecasting setting:** The framework only uses information available before scheduling for evaluation, particularly day-ahead weather forecasts. Its effectiveness is examined through rolling-origin evaluation, day-type analysis, encoding-strategy comparison, ablation study, and SHapley Additive exPlanations (SHAP)-based interpretation.

2. Proposed Methodology

This section describes the theoretical basis, authorial design, and implementation logic of the proposed CA-FDF framework. Instead of relying on a fully end-to-end black-box strategy, CA-FDF separates the main sources of substation load variation into several connected modules. As shown in Figure 1, the workflow includes an input layer, a base branch, a residual refinement branch, a final fusion-output module, and a SHAP-based interpretability module. In this design, Discrete Wavelet Transform (DWT)-based fre-

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quency decoupling and Recursive Least Squares (RLS) tracking are used to estimate the low-frequency load base, calendar-state encoding and lagged residual learning are used to correct institutional schedule shifts, and Multi-Layer Perceptron (MLP)-enhanced weather representation is used to refine the remaining weather-sensitive residual variation. Therefore, the own contribution of CA-FDF is a staged residual-correction framework that assigns different forecasting errors to corresponding model components, rather than forcing a single end-to-end model to learn all effects at once.

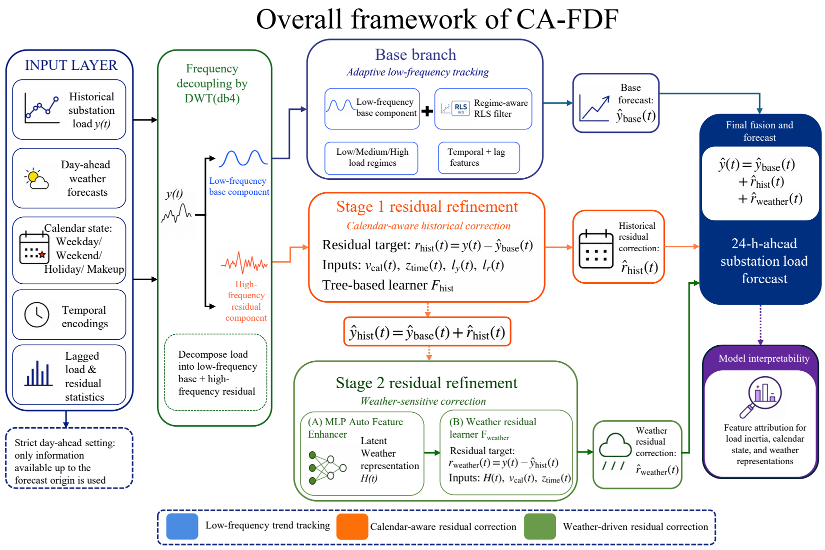

Figure 1. Overall framework of CA-FDF. The framework consists of an input layer, a DWT-RLS-based base branch, a residual refinement branch with calendar-aware and weather-sensitive correction, a final fusion-output module, and a SHAP-based interpretability module. The low-frequency base estimate and the refined residual corrections are combined to produce the final day-ahead load forecast. Source: Authors' own elaboration.

2.1. Frequency Decoupling and Adaptive Base Tracking

The original substation load data are unstable, with slow-moving trends and short-term fluctuations mixed together. To separate these two parts, the original load signal $y(t)$ is first decomposed using DWT. In the rolling-origin implementation, this decomposition is recomputed at each forecasting step using only observations available up to the forecast origin. The Daubechies 4 (db4) mother wavelet is chosen because its orthogonal structure is suitable for representing step-like changes and minimizes the distortion at the boundaries. At decomposition level J, the signal is written as

$$y(t) = \sum_k cA_{J,k}\phi_{J,k}(t) + \sum_{j=1}^{J}\sum_k cD_{j,k}\psi_{j,k}(t), \quad (1)$$

where $cA_{J,k}$ and $cD_{j,k}$ are the approximation and detail coefficients, respectively.

The low-frequency component reconstructed from the approximation coefficients is denoted by $y_{base}(t)$. It represents the slowly varying background load and is tracked by an RLS filter. Let $\mathbf{x}(t)$ denote the regressor vector formed by temporal features and lag-based historical statistics. In the implementation, this vector includes cyclical hour-of-day, day-of-week, and day-of-year encodings, together with recent and weekly load-lag statistics available before the forecast origin. These inputs are used only for tracking the low-frequency base component, rather than directly predicting the final load. The weighted least-squares objective is

$$E(t) = \sum_{i=1}^{t} \lambda^{t-i} \left[y_{base}(i) - \boldsymbol{\theta}^T(t)\mathbf{x}(i) \right]^2, \tag{2}$$

where $\lambda \in (0, 1]$ is the forgetting factor and $\boldsymbol{\theta}(t)$ is the coefficient vector. A smaller λ gives more weight to recent observations and therefore allows for faster adaptation to structural change, while a larger λ leads to smoother base tracking. The recursive update equations are

$$\mathbf{K}(t) = \frac{\mathbf{P}(t-1)\mathbf{x}(t)}{\lambda + \mathbf{x}^T(t)\mathbf{P}(t-1)\mathbf{x}(t)}, \tag{3}$$

$$\boldsymbol{\theta}(t) = \boldsymbol{\theta}(t-1) + \mathbf{K}(t) \left[y_{base}(t) - \boldsymbol{\theta}^T(t-1)\mathbf{x}(t) \right], \tag{4}$$

$$\mathbf{P}(t) = \frac{1}{\lambda} \left[\mathbf{I} - \mathbf{K}(t)\mathbf{x}^T(t) \right] \mathbf{P}(t-1), \tag{5}$$

where $\mathbf{K}(t)$ denotes the Kalman gain vector and $\mathbf{P}(t)$ denotes the inverse correlation matrix.

To retain adaptability under different operating conditions, base tracking is further carried out in a regime-aware manner. In practice, the historical load distribution is divided into low-, medium-, and high-load regimes according to empirical quantiles, and the routed base estimates are then recombined into the final low-frequency trend estimate. This regime-aware treatment improves local adaptability while keeping the overall frequency-decoupled structure unchanged. The output of this branch is denoted by $\hat{y}_{base}(t)$ and is used as the background forecast for the subsequent residual-refinement stages.

Figure 2 shows an example of load data decomposition for a representative anonymized substation by using DWT. The low-frequency part keeps the main trend and the smoother daily shape of the original data, while the value of the residual component is relatively small and mainly contains short-term changes. This is also consistent with the design of the framework. The base branch is used to follow the stable part of the load series, and the residual branches are used to capture the more sudden changes that may be related to calendar pattern shifts or weather effects. The low-frequency curve shown in Figure 2 is only an example of the decomposition result from the observed series, not the forecast output itself. In actual prediction, the base branch tracks this low-frequency structure to obtain $\hat{y}_{base}(t)$, as the main part of the final prediction.

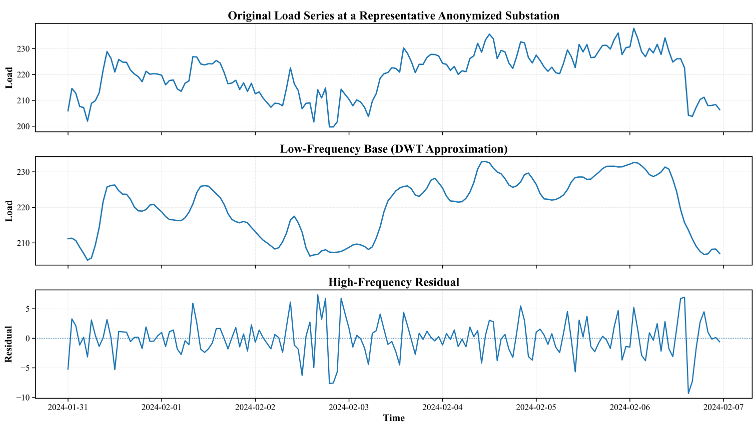

Figure 2. Example of DWT-based load decomposition of an anonymized substation. The top panel presents the original load data, the middle panel reconstructs the low-frequency base, and the bottom panel shows the high-frequency residual component. Source: Authors' own calculation based on anonymized operational load data.

2.2. Knowledge-Driven Calendar-State Encoding and First-Stage Residual Refinement

Although the DWT-RLS branch captures the continuous basic inertia of the load data, high-frequency residuals often show sharp and discrete shifts driven by human activity. In China, substation loads are particularly influenced by the "makeup workday" policy, which can lead to workday-like peaks on nominal weekends. To account for these effects and avoid ordinal bias from numerical coding, we define a categorical operational status indicator $I_{op}(t)$:

$$I_{op}(t) = \begin{cases} \text{Holiday}, & \text{if } t \in \mathcal{H}, \\ \text{Makeup}, & \text{if } t \in \mathcal{M}, \\ \text{Weekend}, & \text{if } t \in \mathcal{W}_{end}, \\ \text{Weekday}, & \text{otherwise}, \end{cases} \tag{6}$$

where $\mathcal{H}$, $\mathcal{M}$, and $\mathcal{W}_{end}$ denote statutory holidays, makeup dates, and regular weekends, respectively.

This categorical state is converted into a four-dimensional one-hot vector:

$$\mathbf{v}_{cal}(t) = [v_H(t), v_M(t), v_{WE}(t), v_{WD}(t)]^T \in \{0,1\}^4. \tag{7}$$

For example, for a makeup workday, $\mathbf{v}_{cal}(t) = [0,1,0,0]^T$.

Let the preliminary prediction produced by the base branch be denoted by $\hat{y}_{base}(t)$. The first-stage residual target is then

$$r_{hist}(t) = y(t) - \hat{y}_{base}(t). \tag{8}$$

This residual captures the nonlinear deviations left after the low-frequency base has been removed. To model it, we combine calendar-state information with temporal encodings, lagged load statistics, and lagged residual cues into a history-aware feature vector:

$$\mathbf{z}_{hist}(t) = \mathbf{v}_{cal}(t) \oplus \mathbf{z}_{time}(t) \oplus \mathbf{l}_y(t) \oplus \mathbf{l}_r(t), \tag{9}$$

where $\mathbf{z}_{time}(t)$ denotes cyclical temporal features such as hour-of-day and day-of-week information, $\mathbf{l}_y(t)$ denotes lagged load features, and $\mathbf{l}_r(t)$ denotes lagged residual features. In the experiments, the cyclical time vector is formed as

$$\mathbf{z}_{time}(t) = [\sin(2\pi h_t/24), \cos(2\pi h_t/24), \sin(2\pi d_t/7), \cos(2\pi d_t/7), \sin(2\pi q_t/366), \cos(2\pi q_t/366)], \tag{10}$$

where h_t, d_t, and q_t denote the hour of day, day of week, and day of year, respectively. The lagged load vector is defined as

$$\mathbf{l}_y(t) = [y(t-24), y(t-48), y(t-168), \overline{y}_{24}(t), \overline{y}_{168}(t), s_y^{24}(t)], \tag{11}$$

and the lagged residual vector is defined as

$$\mathbf{l}_r(t) = [r_{hist}(t-24), r_{hist}(t-48), r_{hist}(t-168), \overline{r}_{24}(t), s_r^{24}(t)]. \tag{12}$$

Here $\overline{y}_{24}(t)$ and $\overline{y}_{168}(t)$ denote the rolling mean of the observed load over the previous 24 and 168 h, $s_y^{24}(t)$ denotes the rolling standard deviation over the previous 24 h, and $\overline{r}_{24}(t)$ and $s_r^{24}(t)$ denote the corresponding rolling mean and standard deviation of the historical residual. The lags of 24, 48, and 168 h are used to represent daily persistence, two-day memory, and weekly periodicity. All these quantities are calculated only from timestamps earlier than t.

The corresponding residual correction is written as

$$\hat{r}_{hist}(t) = \mathcal{F}_{hist}\big(\mathbf{z}_{hist}(t)\big), \tag{13}$$

where $\mathcal{F}_{hist}(\cdot)$ denotes a LightGBM-type Gradient-Boosted Decision Tree (GBDT) residual learner in this study. This model is adopted because it can handle heterogeneous tabular inputs, one-hot calendar variables, and nonlinear interactions between lagged-load and residual features effectively.

It should be noted that $r_{hist}(t)$ is not replaced by $\mathbf{l}_y(t)$ or $\mathbf{l}_r(t)$. Instead, $r_{hist}(t)$ is the supervised target of the first-stage residual learner, while $\mathbf{l}_y(t)$ and $\mathbf{l}_r(t)$ are historical explanatory features used to predict this residual target.

At this stage, the model focuses on distortions tied to work–rest transitions, short-memory dependence, and discrete operating-regime shifts, rather than asking the base branch to absorb all local irregularities. The refined prediction after this stage is

$$\hat{y}_{hist}(t) = \hat{y}_{base}(t) + \hat{r}_{hist}(t). \tag{14}$$

In this way, calendar-aware historical refinement bridges the gap between the smooth low-frequency base and the remaining exogenous fluctuations that are handled in the next stage.

2.3. Cross-Domain Latent Representation Boosting and Weather-Residual Refinement

After the calendar-aware historical correction, the remaining residual may still contain weather-sensitive variation and other short-term deviations. To model the weather-sensitive part, the framework uses day-ahead forecasts of six weather variables: shortwave radiation, 2 m temperature, 2 m relative humidity, 10 m wind speed, precipitation, and cloud cover.

The second-stage residual target is defined as

$$r_{weather}(t) = y(t) - \hat{y}_{hist}(t). \tag{15}$$

This residual represents the part that is still not explained after the base branch and the calendar-aware historical correction.

Using raw weather variables directly in a tree-based model may not be sufficient to represent their joint nonlinear effects. For example, the influence of temperature on load may depend on humidity, cloud cover, solar radiation, and concurrent operating conditions. To better capture such interactions, we introduce an MLP-based AutoFeatureEnhancer. For each forecast timestamp, the weather input vector is defined as

$$\mathbf{W}(t) = [G(t), T(t), RH(t), WS(t), P(t), CC(t)], \tag{16}$$

where $G(t)$ is shortwave radiation, $T(t)$ is 2 m temperature, $RH(t)$ is 2 m relative humidity, $WS(t)$ is 10 m wind speed, $P(t)$ is precipitation, and $CC(t)$ is cloud cover. The MLP maps the weather input into a latent weather representation $\mathbf{H}(t)$ as follows:

$$\mathbf{H}(t) = \mathrm{ReLU}(\mathbf{W}_{h2}\,\mathrm{ReLU}(\mathbf{W}_{h1}\mathbf{W}(t) + \mathbf{b}_1) + \mathbf{b}_2), \tag{17}$$

where $\mathbf{W}_{h1}$ and $\mathbf{W}_{h2}$ are trainable weight matrices, $\mathbf{b}_1$ and $\mathbf{b}_2$ are bias vectors, and $\mathrm{ReLU}(x) = \max(0, x)$ is the rectified linear activation function. The output $\mathbf{H}(t)$ is not a weather forecast itself, but a learned representation of the day-ahead weather inputs for residual load correction.

The learned weather representation is then combined with selected calendar and temporal predictors to estimate the weather-sensitive residual correction:

$$\hat{r}_{weather}(t) = \mathcal{F}_{weather}\big(\mathbf{H}(t) \oplus \mathbf{v}_{cal}(t) \oplus \mathbf{z}_{time}(t)\big), \tag{18}$$

where $\mathcal{F}_{weather}(\cdot)$ denotes the weather residual learner, which is also implemented as a LightGBM-type GBDT model in this study. This learner uses the MLP-derived weather representation, together with calendar and temporal predictors, to estimate the remaining residual after base tracking and calendar-aware historical correction. The final forecast is therefore written as

$$\hat{y}(t) = \hat{y}_{base}(t) + \hat{r}_{hist}(t) + \hat{r}_{weather}(t). \tag{19}$$

The base branch provides the smooth background trend, the first residual stage handles deviations related to calendar effects and load history, and the weather stage further corrects the remaining exogenous variation. These three parts are designed to work in sequence, with each stage focusing on a different source of forecasting error.

2.4. Model Interpretability via SHAP Analysis

To improve the transparency of the proposed CA-FDF for operational use, we integrate SHapley Additive exPlanations (SHAP) based on cooperative game theory, which provides a unified framework for interpreting the predictions of complex machine learning models by assigning an importance value to each feature for a particular prediction. The SHAP value ϕ_i for feature i is calculated as

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!}[v(S \cup \{i\}) - v(S)], \tag{20}$$

where F is the set of all input features, S is a subset of features excluding i, and $v(S)$ represents the expected model output given the feature subset S. By computing these values across the test set, we quantify both the global importance and the local effects of calendar-state variables and learned weather representations. Since several meteorological inputs may be correlated in practice, such as temperature, humidity, cloud cover, and solar radiation, the resulting SHAP values are used to explain the behavior of the trained model rather than to claim independent causal effects of individual variables. In other words, a high SHAP contribution indicates that the model relies strongly on the corresponding feature or feature representation under the given input distribution, but the attribution should still be interpreted together with domain knowledge and the correlations among weather variables.

3. Case Study and Results

3.1. Data Description and Evaluation Setup

The proposed framework is evaluated on hourly operational load data from 29 representative 220 kV substations in a provincial power grid in East China for the year 2024. As 2024 was a leap year, each substation has 8784 hourly observations. The substations we selected have different load characteristics. Some are mainly industrial areas, while others are mixed commercial and residential areas. This makes the test results more closely reflect practical dispatch scenarios.

Because the original station-level operational records are subject to utility confidentiality restrictions, the raw load values and any identifiable station or geographic information cannot be publicly released. To clarify the data structure used in the experiments, Table 1 summarizes the main data sources, their key fields, and how they are linked before model training. The dataset is organized as an hourly panel indexed by anonymized substation ID and timestamp. After preprocessing and feature construction, each model sample contains the observed active load as the forecasting target, the calendar-state label, the matched day-ahead weather forecast variables, and lagged or rolling historical features calculated only from earlier timestamps.

Table 1. Data structure and linkage of the experimental dataset. All substation identifiers and geographic labels are anonymized. Source: Authors' own elaboration.

Data Source	Key Fields	Temporal Resolution/Size	Role in the Experiment
Hourly substation load records	Anonymized substation ID; timestamp; hourly active load	29 anonymized substations × 8784 hourly observations in 2024	Forecasting target and source of lagged-load features
Substation–weather-zone mapping	Anonymized substation ID; anonymized weather-zone code	One anonymized weather-zone code for each substation	Links each substation to the corresponding weather forecast records without disclosing geographic names
Day-ahead meteorological forecast records	Timestamp; shortwave radiation; 2 m temperature; 2 m relative humidity; 10 m wind speed; precipitation; cloud cover	Hourly forecast records for the study period	Exogenous inputs for the weather-residual refinement branch
Derived calendar and history features	Calendar state; lagged load; lagged residual; rolling statistics	Calculated for each forecast timestamp using only historical information	Inputs for calendar-aware and history-aware residual learning

The continuous variables used after data linkage and feature construction are further summarized in Table 2, including their physical units and descriptive statistics. The load variable is hourly active load, measured in MW. The meteorological variables are day-ahead forecast inputs used by the weather-residual branch. Calendar-state variables are dimensionless categorical indicators constructed from the official Chinese holiday and makeup-workday schedule, and all lagged and rolling features are calculated only from timestamps earlier than the forecast target.

Table 2. Descriptive statistics of the main continuous variables used in the experiments. Source: Authors' own calculation.

Variable	Unit	Mean	Std.	Min.	Max.
Hourly active load	MW	186.88	100.20	12.81	529.07
Shortwave radiation	W/m^2	156.30	242.29	0.00	1046.00
2 m temperature	°C	18.71	9.37	−5.70	41.61
2 m relative humidity	%	75.89	16.78	18.00	100.00
10 m wind speed	m/s	7.58	5.44	0.00	44.74
Precipitation	mm	0.17	0.75	0.00	34.70
Cloud cover	%	70.69	36.52	0.00	100.00

Before modeling, only isolated and physically implausible spikes that were inconsistent with the neighboring load profile were capped using a rolling 3σ rule. This treatment was intended to reduce the influence of likely sensor or recording errors, not to remove real operational load anomalies. Persistent deviations associated with holidays, makeup workdays, weather changes, or load-regime transitions were retained and modeled through the calendar-state, weather, and lagged residual features. Missing values were filled by linear interpolation using only information available before the forecast origin. No samples from the target day or later were used during pre-processing.

To reflect changing seasonal conditions and keep the evaluation close to actual operation, we use an expanding-window rolling forecast scheme instead of a fixed train–test split. For each target day T, the data are divided in chronological order into four non-overlapping parts:

- **Testing Set:** The target day T (24 h), used only for out-of-sample day-ahead prediction.
- **Validation Set:** The 7 days immediately before the target day (days $T-7$ to $T-1$), used for hyperparameter selection in the current rolling step.
- **Isolation Gap**: A 24 h gap at day $T-8$, kept out of the validation scoring during hyperparameter selection to reduce short-range leakage between model selection and model fitting.
- **Sub-training Set:** All available historical data from day 1 to day $T-9$, used to learn the initial model structure and the long-term load background.

At each rolling step, the forecasting procedure is carried out in two stages. We first search for a suitable hyperparameter configuration on the validation set. After that, the model is retrained using all historical data from day 1 to day $T-1$ under the selected configuration, and the load of day T is then predicted in a strict day-ahead setting. For the benchmark comparison and the calendar-related analysis reported later, all results are summarized on the common evaluation subset, where every compared model produces predictions for the same station–timestamp pairs. In this way, all reported metrics are calculated on exactly the same samples.

The parameter ranges in Table 3 were selected to balance model flexibility and forecasting stability under the rolling-origin validation scheme. The DWT decomposition level was limited to 2–4, because a very shallow decomposition may not sufficiently separate the base and residual components, while an overly deep decomposition may move useful load variation into the residual branch. The RLS forgetting factor was searched within 0.95–0.995 to balance fast adaptation to recent changes and stable low-frequency tracking. For the GBDT-based residual learners, the learning rate and tree-complexity parameters were restricted to moderate ranges to capture nonlinear residual patterns without overfitting the short validation window. The MLP weather enhancer used a compact two-hidden-layer structure, together with L2 regularization, so that the learned weather representation remained stable and did not dominate the calendar-aware residual correction. At each rolling step, the final hyperparameter configuration was selected on the validation set and then used for the corresponding day-ahead test prediction.

Table 3. Hyperparameter configurations and search space of the CA-FDF framework. Source: Authors' own elaboration.

Module	Hyperparameter	Base Value	Search Space
FDF: DWT + RLS	Mother Wavelet	Daubechies 4 (db4)	-
	Decomposition Level (J)	2	[2, 4], integer
	RLS Forgetting Factor (λ)	0.98	[0.95, 0.995], log-uniform
Historical Residual Refinement	Number of Estimators	200	-
	Learning Rate	0.05	[0.01, 0.2], log-uniform
	Number of Leaves/Tree Complexity	31	[20, 50], integer
MLP Weather Enhancer	Hidden Layer Sizes	(32, 16)	-
	L2 Regularization (α)	0.001	$[10^{-5}, 10^{-1}]$, log-uniform
Weather Residual Learner	Number of Estimators	150	-
	Maximum Tree Depth	5	[3, 8], integer
	Learning Rate (η)	0.05	[0.01, 0.2], log-uniform

All experiments were implemented in Python 3.11 under the same rolling-origin evaluation protocol. In CA-FDF, the DWT decomposition, regime-aware RLS base tracking, LightGBM-type GBDT residual learners, and MLP-based weather representation were executed sequentially for each rolling forecast step. For the baseline models, including DLinear, PatchTST, and CNN-BiLSTM-Attention, the main architectural settings were kept consistent with their corresponding reference designs, and tunable training hyperparameters were selected using the same validation window as the proposed CA-FDF framework. No test-set information was used for model selection. The experiments were conducted on a Windows-based workstation equipped with an Intel Core i7 CPU and an NVIDIA GeForce RTX 4060 GPU. All reported metrics were calculated on the same evaluation samples after aligning the substation IDs and timestamps across all models, so that different models were compared using exactly the same observations.

3.2. Evaluation Metrics

To assess forecasting accuracy, four standard metrics are employed: Mean Absolute Percentage Error (MAPE), Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and R-squared (R^2). Let y_i and $\hat{y}_i$ denote the observed and predicted load values at sample i, respectively, N denote the number of evaluated samples, and $\bar{y}$ denote the mean observed load. The four metrics are defined as follows:

$$\mathrm{MAPE} = \frac{100\%}{N}\sum_{i=1}^{N}\left|\frac{y_i - \hat{y}_i}{y_i}\right|, \tag{21}$$

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|, \tag{22}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}, \tag{23}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2}. \tag{24}$$

MAPE measures the relative forecasting error and is useful for comparing substations with different load scales. MAE and RMSE measure the absolute forecasting error in MW, while RMSE gives a larger penalty to large deviations and is therefore more sensitive to peak or valley forecasting errors. The R^2 value measures how much of the variation in the observed load can be explained by the forecast. In this study, the four metrics are reported together to provide a balanced evaluation of both relative accuracy and absolute error magnitude.

3.3. Benchmark Comparison

To assess the forecasting performance of CA-FDF, we compare it with three representative baseline models under the same 24 h ahead rolling-origin setting. All models are evaluated with the same forecasting horizon and the same evaluation protocol so that the comparison is fair.

- **PatchTST [18]:** A Transformer-based model that uses patching and channel-independent design to capture long-range temporal dependence.
- **DLinear [19]:** A linear decomposition model that has shown competitive performance in many time-series forecasting tasks.
- **Optimized CNN-BiLSTM-Attention [11]:** A hybrid sequential model commonly used in power load forecasting.

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As shown in Table 4, CA-FDF gives the best results on the aligned evaluation set of 29 substations. It achieves the lowest MAPE (1.92%), RMSE (5.26), and MAE (3.40), and also obtains the highest R^2 (0.9972). This shows that the proposed framework predicts substation-level load more accurately in the current test setting.

Table 4. Performance comparison on the aligned evaluation set covering 29 substations. Source: Authors' own calculation.

Model Architecture	MAPE	RMSE (MW)	MAE (MW)	R^2
DLinear [19]	2.33%	5.67	3.90	0.9968
Optimized CNN-BiLSTM-Attention	2.50%	6.15	4.26	0.9962
PatchTST [18]	2.53%	7.67	4.62	0.9941
Proposed Framework (CA-FDF)	**1.92%**	**5.26**	**3.40**	**0.9972**

The overall performance of DLinear is the strongest among the three baselines, but CA-FDF still improves on it, reducing MAPE from 2.33% to 1.92% and MAE from 3.90 to 3.40. The difference is larger when compared with PatchTST. The same general tendency can also be seen in the 72 h forecasting example, the error-distribution results, and the calendar-aware analysis shown later. This indicates that splitting the forecast into a low-frequency base part and later residual corrections helps reduce the over-smoothing problem of end-to-end sequence models.

3.4. Visual Error and Over-Smoothing Analysis

To further clarify the over-smoothing issue in end-to-end models, we first examine a representative 72 h forecasting example and then compare the distribution of absolute errors across all substations.

As shown in Figure 3, all models capture the general trend of the load series, but their performance differs when the load changes quickly. The comparison is shown over a continuous 72 h forecasting window at a representative substation.

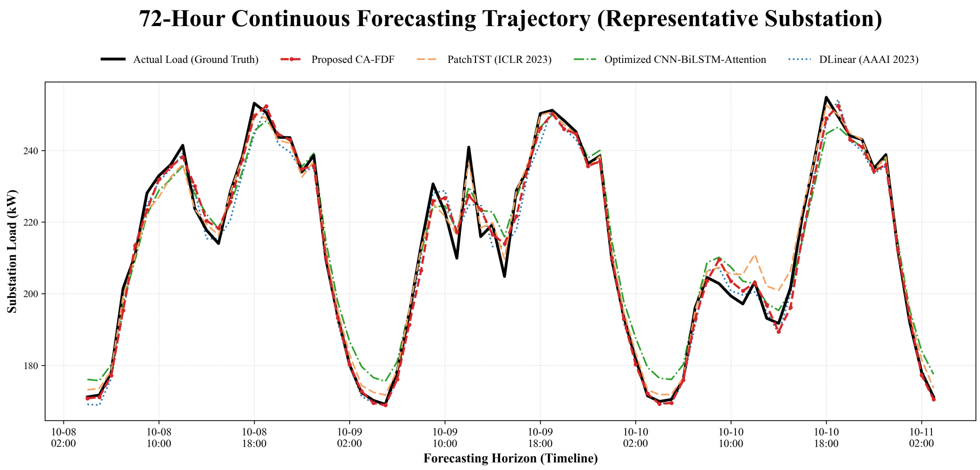

Figure 3. The 72 h continuous forecasting trajectory at a representative substation. Source: Authors' own calculation based on the aligned evaluation set.

The baseline models, including DLinear, CNN-BiLSTM-Attention, and PatchTST, respond more smoothly around peak and valley transitions, especially during the sharp increase in the evening load or the low demand in the early morning. By comparison, CA-FDF follows the local fluctuations more closely and fits the actual load data better at these turning points. This indicates that the proposed frequency-decoupled and residual-refinement structure enables the model to capture short-term changes more effectively.

Figure 4 shows a similar situation from the perspective of error distribution. The error distributions of the three baselines are wider and have heavier upper tails, while for the proposed model, most of the absolute errors are concentrated in a smaller numerical range. The dashed markers inside each violin summarize the central part of the distribution. Overall, CA-FDF shows a lower central tendency and a tighter spread on the aligned test set.

To further examine whether the improvement is statistically reliable and not caused by a single representative substation, we performed a station-level paired significance test. For each of the 29 substations, the MAE of each model was first calculated on the aligned evaluation period. The proposed CA-FDF and each baseline were then compared using a two-sided Wilcoxon signed-rank test over the 29 paired station-level MAE values. A significance level of 0.05 was used to assess whether the station-level MAE reduction was statistically significant. The results are reported in Table 5.

Table 5. Station-level statistical comparison between CA-FDF and baseline models based on MAE. Source: Authors' own calculation.

Comparison	p-Value	Stations with Lower MAE
CA-FDF vs. DLinear	1.84×10^{-5}	24/29
CA-FDF vs. CNN-BiLSTM-Attention	3.37×10^{-6}	25/29
CA-FDF vs. PatchTST	3.73×10^{-8}	27/29

The test results show that the reduction in station-level MAE is statistically significant for all three baseline comparisons. In addition, CA-FDF obtains lower MAE than DLinear, CNN-BiLSTM-Attention, and PatchTST at 24, 25, and 27 out of the 29 substations, respectively. This indicates that the improvement is not only visible in the representative 72 h trajectory, but is also consistent across most substations.

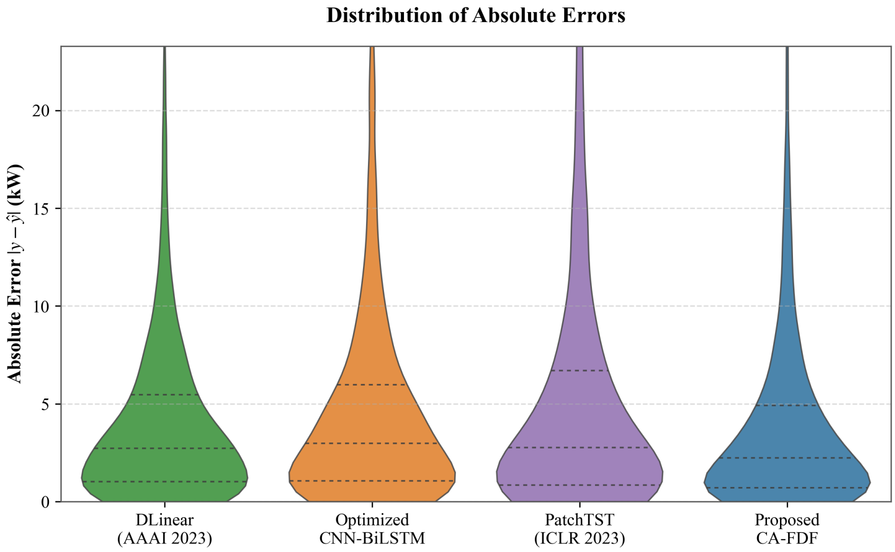

Figure 4. Violin plot of absolute forecasting errors on the aligned test set for DLinear, CNN-BiLSTM-Attention, PatchTST, and the proposed CA-FDF. The proposed model shows more errors concentrated at lower values than the three baselines. Source: Authors' own calculation.

3.5. Calendar-Aware Evaluation Across Day Types

To test the effect of calendar-state encoding, we further compare the forecasting results of four operational day types: Weekday, Weekend, Holiday, and Makeup. This analysis focuses on two questions. The first is whether the load patterns of these day types are different in the raw data. The second is whether distinguishing them explicitly is helpful for forecasting under calendar-sensitive conditions.

Figure 5 shows the average daily load curves of a representative anonymized substation under the four day types. The differences are fairly clear. Weekday and Weekend follow a similar daily shape, but the Weekday curve stays at a higher level for most hours. Holiday days have a lower overall load level and a more pronounced midday trough, which makes them different from ordinary non-working days. Makeup days show the highest load level among the four categories and remain above ordinary Weekend and Holiday conditions for most of the day. This suggests that makeup workdays should not be grouped together with regular weekends, and supports treating them as a separate calendar state.

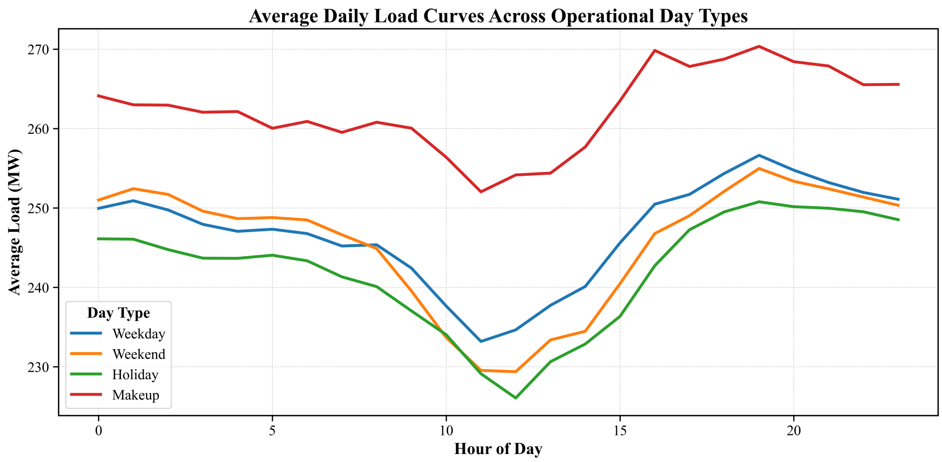

Figure 5. Average daily load curves across Weekday, Weekend, Holiday, and Makeup conditions at a representative anonymized substation. The four curves show noticeable differences, especially between Holiday and Makeup conditions, which supports the use of explicit calendar-state modeling. Source: Authors' own calculation based on anonymized operational load data.

After showing that the four operational states are associated with different load patterns, we next examine whether the proposed framework can use this information in forecasting. Table 6 reports the daily averaged MAPE of each model across the four day types, and Figure 6 presents the absolute results, together with the relative reduction achieved by the proposed model.

Table 6. Calendar-aware evaluation across different operational day types (daily averaged MAPE, %). Source: Authors' own calculation.

Day Type	Proposed	DLinear	CNN-BiLSTM-Attention	PatchTST
Weekday	**1.91**	2.34	2.51	2.34
Weekend	**1.88**	2.30	2.53	3.19
Holiday	**1.98**	2.36	2.42	2.48
Makeup	**2.02**	2.17	2.46	2.40

As shown in Table 6, CA-FDF achieves the lowest daily averaged MAPE for all four day types. Its advantage is not limited to ordinary weekdays, and can also be seen on holiday and makeup days, where load behavior is more sensitive to calendar arrangement.

Figure 6 shows the same pattern in a more direct way: the upper panel reports the absolute daily averaged MAPE across day types, while the lower panel shows the relative MAPE reduction of the proposed model compared with each baseline. The improvement remains visible across all four day types, including the more irregular holiday and makeup cases.

Taken together, Figures 5 and 6 and Table 6 suggest the same general pattern. Different calendar states are associated with different demand characteristics, and distinguishing them explicitly is helpful for forecasting. These results suggest that calendar-state modeling improves performance under calendar-sensitive and atypical operating conditions.

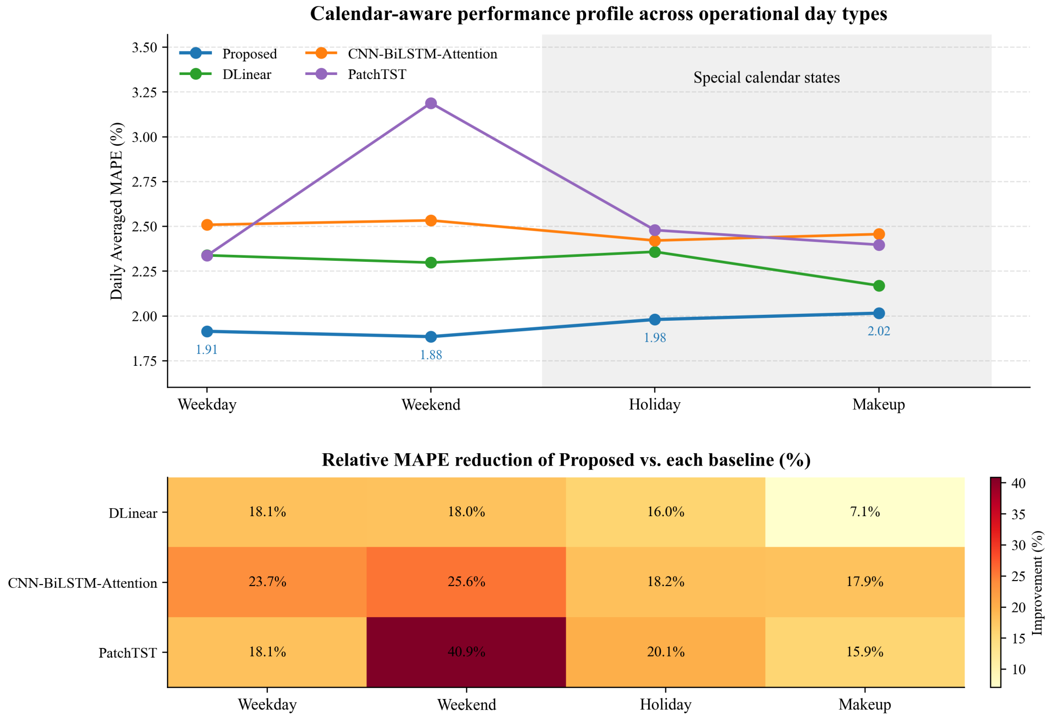

Figure 6. Calendar-aware evaluation across four operational day types. The upper panel shows the daily averaged MAPE of each model under weekday, weekend, holiday, and makeup conditions. The lower panel reports the relative MAPE reduction of the proposed method compared with each baseline. Holiday and makeup are highlighted as the most calendar-sensitive operating states. Source: Authors' own calculation.

3.6. Effect of Calendar Encoding Strategy

To examine the role of the calendar encoding design, we compare three calendar encoding settings under the same rolling-origin evaluation setting. C1 uses only conventional cyclical time encoding, including hour-of-day, day-of-week, and day-of-year periodic features. C2 adds a binary holiday indicator to C1. C3, which is the setting used in the proposed CA-FDF framework, further replaces the binary flag with an explicit calendar-state encoding that distinguishes weekday, weekend, holiday, and makeup conditions through one-hot representation.

Table 7 shows that C3 gives the strongest overall performance among the three settings. Compared with C1, C3 reduces MAPE from 1.93% to 1.92%, while also lowering RMSE from 5.29 to 5.26 and MAE from 3.43 to 3.40. The numerical gap between C3 and C1 is small, but the direction is consistent across all four metrics, including a slight increase in R^2 from 0.9969 to 0.9972.

Table 7. Comparison of different calendar encoding strategies on the full test set. Source: Authors' own calculation.

Encoding Setting	MAPE	RMSE (MW)	MAE (MW)	R^2
C1: Conventional Time Encoding Only	1.93%	5.29	3.43	0.9969
C2: C1 + Binary Holiday Flag	1.96%	5.39	3.49	0.9961
C3: Explicit Calendar-State Encoding (Proposed)	**1.92%**	**5.26**	**3.40**	**0.9972**

The performance of C2 is worse than that of both C1 and C3. This suggests that a simple binary holiday flag is not sufficient to describe operational schedule changes at the substation level. In practice, such a flag only separates holidays from non-holidays, without explicitly distinguishing ordinary weekends and makeup workdays. This makes it harder for the model to tell these different day types apart. By contrast, the explicit calendar-state encoding used in C3 keeps the distinction among weekday, weekend, holiday, and makeup conditions, which allows the residual branch to respond more appropriately to different demand patterns.

This result is consistent with the day-type analysis in Figure 5. Since holiday and makeup conditions show visibly different load profiles from ordinary weekends, treating them as separate operational states is more in line with the observed data than using a single binary holiday label. Overall, the C1–C3 comparison suggests that the gain from the proposed framework is related not only to adding calendar information, but also to representing that information in a more suitable way.

3.7. SHAP Feature Attribution and Interpretability Analysis

A common limitation of high-accuracy forecasting models in power system applications is limited interpretability. After examining the predictive results and the calendar-aware behavior of the model, we further analyze whether its feature usage is consistent with domain intuition through SHAP. Specifically, a station-level global SHAP analysis is conducted on an anonymized representative substation, and the resulting beeswarm plot is shown in Figure 7.

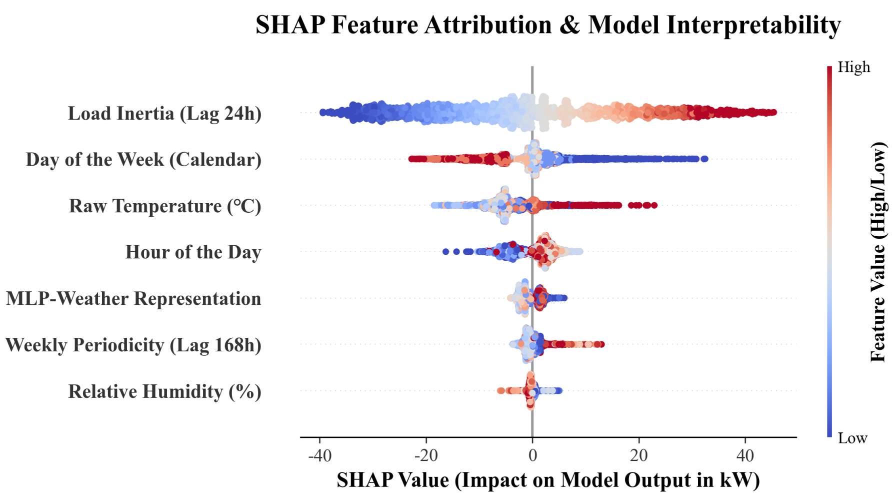

Figure 7. SHAP beeswarm plot for feature attribution analysis. The horizontal axis represents the impact on the model output (SHAP value), while the color gradient indicates the magnitude of the original feature value (red for high, blue for low). Source: Authors' own calculation.

The SHAP results show a clear ordering in the factors associated with the forecast. *Load Inertia (Lag 24 h)* remains the most influential feature, as reflected by the widest SHAP spread in the beeswarm plot. Larger lagged-load values generally push the prediction upward, while smaller lagged-load values contribute negatively. This pattern is consistent with the role of the DWT-RLS base branch, which is designed to track the low-frequency inertia in substation load evolution.

After this autoregressive effect is taken into account, the influence of *Day of the Week (Calendar)* feature is still quite visible. Its SHAP range is wider than that of the weather-related features, which suggests that calendar information remains an important source of residual variation. The bidirectional spread also indicates that different calendar states may either raise or lower the predicted load, depending on the operating regime. This is consistent with the motivation of the proposed framework: weekends, holidays, and makeup workdays reflect institutional schedule changes rather than ordinary temporal positions, so their effects are difficult to capture through generic timestamp features alone.

Weather-related information also contributes to the forecast. Among all the meteorological variables, *Raw Temperature (°C)* shows the most obvious influence, indicating that thermal conditions are still an important external driver of local demand fluctuations. At the same time, the *MLP-Weather Representation* feature also contributes, which suggests that the learned latent embedding captures more complex nonlinear weather effects that are not present in the raw variables. In this sense, the weather branch appears to refine the forecast after the main structure related to inertia and calendar has been established.

A similar reading applies to *Hour of the Day* and *Weekly Periodicity (Lag 168 h)*; both provide moderate and stable contributions related to intraday rhythm and recurring weekly dependence. By contrast, *Relative Humidity (%)* plays a smaller role in this representative case. Overall, the SHAP analysis is in line with the staged design of CA-FDF: the forecast is first determined by recent load inertia, then adjusted according to calendar regime, and finally refined through weather-sensitive corrections. This ordering matches the attribution pattern in Figure 7 and supports the use of explicit calendar-state modeling in substation-level load forecasting.

3.8. Ablation Study: Quantifying the Contribution of Key Components

To clarify the contribution of the main components in CA-FDF, we conduct an ablation study under the same rolling-origin evaluation framework. The results are all listed in Table 8. M1 is the DWT-RLS base branch only. M2 adds the calendar-aware historical residual refinement. M3 further adds raw day-ahead weather inputs. M4 is the full CA-FDF model with the MLP-based weather representation.

Table 8. Ablation study: impact of key components on forecasting accuracy. The downward arrows in the MAPE Reduction column indicate relative reductions in MAPE. Source: Authors' own calculation.

Model Version	**MAPE (%)**	**RMSE (MW)**	**MAE (MW)**	**MAPE Reduction**
M1: FDF Base (DWT + RLS Only)	5.86	13.32	9.59	-
M2: M1 + Calendar-Aware Residual Refinement	1.97	**5.25**	3.43	↓ 66.4% (vs. M1)
M3: M2 + Raw Weather Inputs	1.96	5.34	3.48	↓ 0.5% (vs. M2)
M4: Full CA-FDF (M3 + MLP-Weather)	**1.92**	5.26	**3.40**	**↓ 2.0% (vs. M3)**

The result of the ablation study shows that the improvement from M1 to M2 is the most significant, with MAPE dropping from 5.86% to 1.97%. This indicates that frequency decoupling and adaptive base tracking alone are not enough when substation load is affected by discrete schedule changes such as holidays and makeup workdays. After adding the calendar-aware historical residual branch, the error decreases sharply, which suggests that this part of the framework is the main source of the improvement.

When the raw weather data of the next day was added to M3, there was only a very small reduction in MAPE, from 1.97% to 1.96%. At the same time, RMSE and MAE became slightly worse, increasing from 5.25 to 5.34 and from 3.43 to 3.48. This indicates that the historical residual module has already explained most of the variation, and adding raw meteorological inputs alone brings only a marginal MAPE improvement while worsening RMSE and MAE. In other words, weather information is still relevant, but its effect is not fully captured when it is added only in raw form.

Finally, the full model M4 improves MAPE and MAE relative to M3, reducing them to 1.92% and 3.40, while RMSE also improves relative to M3 from 5.34 to 5.26. At the same time, M4 does not outperform M2 on RMSE, since M2 records the lowest RMSE value of 5.25 in Table 8. This means that the learned weather representation helps produce the best overall balance in MAPE and MAE. Combined with the C1–C3 results in Table 7, it is clear that the effectiveness of CA-FDF comes from the way calendar and weather information are organized and used in different stages.

4. Discussion and Limitations

The experimental results indicate that the main advantage of CA-FDF comes from organizing the forecasting task into a low-frequency base branch and sequential residual-refinement stages. The benchmark comparison shows that the proposed framework improves the overall forecasting accuracy, while the ablation study further suggests that the calendar-aware historical residual branch is the dominant source of improvement. The SHAP analysis is also consistent with this staged interpretation, since recent load inertia has the strongest influence, followed by calendar-related features and weather-sensitive corrections.

At the same time, the calendar-aware results should be interpreted with caution. Calendar-aware modeling does not necessarily imply that the largest relative improvement must occur on every special calendar state. Although holiday and makeup days are more calendar-sensitive, their sample sizes are much smaller than ordinary weekday and weekend conditions, which makes the estimated relative improvement more sensitive to station-specific variation. In addition, makeup days are not purely anomalous non-working days; they are administratively shifted working days and therefore often retain a workday-like load pattern. As a result, a strong decomposition-based baseline such as DLinear can already capture part of this behavior through recent and weekly historical information. This explains why the gain over DLinear on Makeup days is modest, even though CA-FDF still achieves the lowest MAPE for this day type. Therefore, explicit calendar-state encoding should be understood as a way to provide a clearer operational distinction among weekday, weekend, holiday, and makeup conditions, rather than as a guarantee of the largest percentage reduction on each special day type.

This study also has several limitations. First, the experiments are based on one year of hourly data from 29 anonymized 220 kV substations in East China. Although the substations cover different load characteristics, the results may still be affected by the specific regional calendar arrangements, weather conditions, and industrial structure in the study area. Second, the proposed framework is evaluated under a strict day-ahead setting, but its reliability under rare extreme weather events, sudden operational changes, or unusual industrial interruptions requires further validation.

The proposed framework is transferable in structure, but not necessarily plug-and-play across all systems. When applied to other regions or countries, the calendar-state definition should be adjusted according to local holiday and working-day arrangements, and the weather variables and hyperparameter ranges should be recalibrated using local validation data. Therefore, the reported results should be interpreted as evidence for the effectiveness of the proposed design under the current regional dataset and day-ahead evaluation setting, rather than as a fully general guarantee across all power-system contexts.

5. Conclusions

This paper proposes the CA-FDF framework for substation-level day-ahead load forecasting under over-smoothing and non-stationary operating conditions. The framework decomposes the load sequence into a low-frequency base component and a high-frequency residual component, and combines explicit calendar-state encoding with staged residual refinement and MLP-enhanced weather representations. In the benchmark comparison with DLinear, PatchTST, and CNN-BiLSTM-Attention, CA-FDF achieves the lowest forecasting error on operational data from 29 substations, with an MAPE of 1.92%. Compared with DLinear, the strongest baseline in terms of MAPE, CA-FDF reduces the MAPE from 2.33% to 1.92%, corresponding to a relative reduction of approximately 17.6%. The calendar-aware evaluation further shows that the advantage of the proposed framework is visible across weekday, weekend, holiday, and makeup conditions. The comparison of calendar encoding settings suggests that explicit calendar-state encoding is more effective than conventional time encoding alone or a binary holiday flag. The decomposition example and the average daily load curves across day types also show that the proposed design is consistent with the observed structure of the data. In addition, the SHAP analysis supports the interpretability of the proposed framework by showing the roles of load inertia, calendar state, and weather-related information. The ablation study further shows that the largest gain comes from the calendar-aware residual branch. The learned weather representation improves the final model mainly in MAPE and MAE relative to the raw-weather variant, while its RMSE remains close to the best ablation result.

Future work can extend the framework in two directions. First, spatial dependence between substations can be incorporated, for example, through graph-based modeling in the low-frequency branch. This direction is motivated by the fact that neighboring or functionally related substations may share regional demand patterns, industrial activity, and weather-driven variations. Modeling such spatial coupling is expected to improve the representation of coordinated load changes across substations. Second, probabilistic forecasting can be developed to quantify uncertainty under extreme weather and special calendar conditions. This extension is motivated by the operational need for risk-aware dispatch under uncertain external conditions, and it is expected to provide prediction intervals or distributional forecasts instead of only point forecasts.

6. Nomenclature and Abbreviations

The main abbreviations and mathematical symbols used in this study are summarized in Tables 9 and 10.

Table 9. Abbreviations used in the manuscript. Source: Authors' own elaboration.

Abbreviation	Meaning
STLF	Short-Term Load Forecasting
CA-FDF	Calendar-Aware Frequency-Decoupled Framework
DWT	Discrete Wavelet Transform
RLS	Recursive Least Squares
MLP	Multi-Layer Perceptron
SHAP	SHapley Additive exPlanations
MAPE	Mean Absolute Percentage Error
RMSE	Root Mean Square Error
MAE	Mean Absolute Error
GBDT	Gradient-Boosted Decision Tree

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Table 10. Main mathematical notation used in the proposed framework. Source: Authors' own elaboration.

Symbol	Meaning	Unit
$y(t)$	Observed active load at time t	MW
$\hat{y}(t)$	Final forecasted active load at time t	MW
$y_{base}(t)$	Low-frequency component reconstructed from DWT approximation coefficients	MW
$\hat{y}_{base}(t)$	Base forecast produced by regime-aware RLS tracking	MW
$r_{hist}(t)$	First-stage residual after base tracking, $y(t) - \hat{y}_{base}(t)$	MW
$\hat{r}_{hist}(t)$	Calendar-aware historical residual correction	MW
$r_{weather}(t)$	Second-stage residual after historical correction, $y(t) - \hat{y}_{hist}(t)$	MW
$\hat{r}_{weather}(t)$	Weather-sensitive residual correction	MW
$\mathbf{v}_{cal}(t)$	One-hot calendar-state vector for holiday, makeup, weekend, and weekday	dimensionless
$\mathbf{z}_{time}(t)$	Cyclical time encoding vector	dimensionless
$\mathbf{l}_y(t)$	Lagged load feature vector	MW
$\mathbf{l}_r(t)$	Lagged residual feature vector	MW
$\mathbf{W}(t)$	Day-ahead weather forecast vector	variable-specific
$\mathbf{H}(t)$	Latent weather representation learned by the MLP	dimensionless
λ	Forgetting factor in the RLS filter	dimensionless

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